

4. Line Plot of Historical Stock Prices Between Two Specific Dates

Aim

To write a Python Pandas program to create a **line plot** of the historical closing stock prices of Alphabet Inc. between two specific dates.

Procedure

1. Import Pandas and Matplotlib libraries.
2. Read the stock dataset using `read_csv()`.
3. Convert the **Date** column into datetime format.
4. Filter the records between the required dates.
5. Plot the **Close** price using a line graph.
6. Display the graph.

Program

```
import pandas as pd
import matplotlib.pyplot as plt

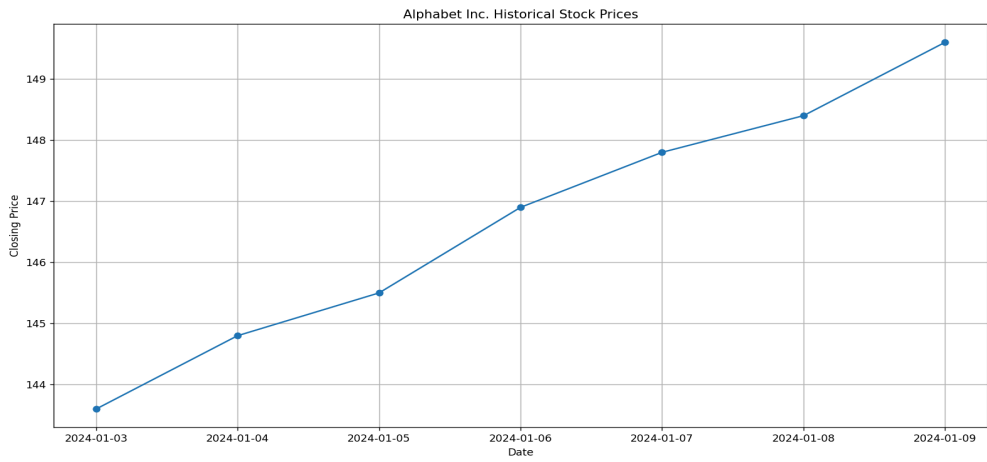
# Read dataset
df = pd.read_csv("alphabet_stock_data.csv")

# Convert Date column
df["Date"] = pd.to_datetime(df["Date"], dayfirst=True)

# Select data between two dates
data = df[(df["Date"] >= "2020-04-01") &
          (df["Date"] <= "2020-04-30")]

# Line Plot
plt.figure(figsize=(10,5))
plt.plot(data["Date"], data["Close"], marker='o')
plt.title("Alphabet Stock Closing Price")
plt.xlabel("Date")
plt.ylabel("Close Price")
plt.grid(True)
plt.show()
```

Figure 1



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5. Bar Plot of Trading Volume Between Two Specific Dates

Aim

To write a Python Pandas program to create a **bar plot** of Alphabet Inc.'s trading volume between two specific dates.

Procedure

1. Import Pandas and Matplotlib.
2. Read the stock dataset.
3. Convert the Date column to datetime format.
4. Filter the required date range.
5. Plot the **Volume** column as a bar chart.
6. Display the graph.

Program

```
import pandas as pd
import matplotlib.pyplot as plt

# Read dataset
df = pd.read_csv("alphabet_stock_data.csv")

# Convert Date
df["Date"] = pd.to_datetime(df["Date"], dayfirst=True)

# Filter dates
data = df[(df["Date"] >= "2020-04-01") &
          (df["Date"] <= "2020-04-30")]

# Bar Plot
plt.figure(figsize=(10,5))
plt.bar(data["Date"].dt.strftime("%d-%m"), data["Volume"])
plt.title("Alphabet Trading Volume")
plt.xlabel("Date")
plt.ylabel("Volume")
plt.xticks(rotation=45)
plt.show()
```

Figure 1

